practical-meteorology-course — User Manual

Contents

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Version 1.0.2 (2026-09-04). Source of truth: this Markdown file; the HTML and PDF beside it are built by `python docs/build_manual.py`. Machine-oriented instructions for AI agents are in `AGENTS.md`.

The material itself is in Portuguese (pt-BR); this manual is in English so that the tooling and the licence conditions are clear to everyone.

1. What this is

Complete teaching material for a two-semester undergraduate course on Roland Stull's open textbook Practical Meteorology: An Algebra-based Survey of Atmospheric Science (v1.02b, UBC, CC BY-NC-SA 4.0). For each of the book's 22 chapters, plus an extra module 23 on climate change (which is not a chapter of the book), it ships five pieces:

Piece	Where	Form
Lecture notes — a blackboard script with `noquadro` boxes, cross-references `\javimos`/`\veremos`, and the chapter's complementary exercises (T1–T5 theory, N1–N4 numerical)	`notas/capNN_notas.tex` + `.pdf`	LaTeX article
Slides — one deck per chapter, every book figure that may be redistributed, plus notebook figures	`slides/capNN_slides.tex` + `.pdf`	Beamer

Piece	Where	Form
Teacher's guide — one section per chapter (`guia.pdf`, 47 pages) and the two-semester schedule (`cronograma.pdf`)	`guia_do_professor/`	LaTeX
Student notebook — four worked cases and the N1–N4 exercise cells, executed	`notebooks/capNN_<tema>.ipynb`	Jupyter
Solutions notebook — T1–T5 checked with SymPy, N1–N4 solved in full, executed	`notebooks/capNN_solucoes.ipynb`	Jupyter

Across the course: `listas/listas_exercicios.pdf` (24 pages, one printable list per chapter, generated from the notes), `indice.pdf` (the course map: overview, the five pieces, chapter table, quick start, licence), and `figuras/capNN/` (the book's figures as JPEG crops plus the PNGs the notebooks generate for the slides).

Start with **`indice.pdf`**.

2. Setup

2.1 Reading and teaching (no tooling)

Every deliverable is committed as a PDF or an executed notebook: clone (or download the release archive) and open them. Nothing to install.

2.2 Running the notebooks

```
conda env create -f environment.yml
conda activate meteo
jupyter lab
```

The `meteo` environment (Python 3.11) carries NumPy, SciPy, SymPy, pandas, xarray, MetPy, Py-ART, Satpy, Cartopy, pint, ipywidgets and, from pip, ambiance, pyrcel, tcpyPI and climlab ≥ 0.9.2. `environment.yml` explains the two pins that matter (matplotlib < 3.11 for MetPy's `StationPlot`; climlab from pip because the conda-forge build breaks with NumPy 2).

Each notebook begins with

```python
import sys; sys.path.append('../comum')
from estilo_meteo import aplicar_estilo, CORES
aplicar_estilo()
```

so run it from `notebooks/` (Jupyter does this when you open the file there).

2.3 Compiling the LaTeX

TeX Live (2026 used here) with `pdflatex`. Every document compiles from its own folder; the shared preamble and Beamer theme live in `comum/` and are found by relative path:

```
cd slides && pdflatex cap05_slides.tex && pdflatex cap05_slides.tex
cd notas  && pdflatex cap05_notas.tex
cd guia_do_professor && pdflatex guia.tex && pdflatex guia.tex
```

Two passes for anything with a table of contents or cross-references.

2.4 Rebuilding figures from the book (maintainers only)

The book is not in the repository. Download the whole-book PDF from https://www.eoas.ubc.ca/books/Practical_Meteorology/ into `book/` (gitignored), then slice → extract → map (section 4). The extractor used was a local MinerU-based pipeline; any extractor that writes `capNN.md` + `capNN_assets/images/<hash>.jpg` works with `map_figs.py`.

3. Using the material

- A chapter in class: notes as the lecturer's script, slides projected, the student notebook opened live for the four cases; the guide section lists the book exercises to assign and the timing.
- Exercises: the printable list (`listas/`) states T1–T5 and N1–N4; the students solve T on paper and N in the empty cells at the end of their notebook; the solutions notebook is for the teacher.
- Schedule: `guia_do_professor/cronograma.pdf` — semester I chapters 1–11, semester II chapters 12–23, 15 weeks each, 2 × 2 h per week, lists due the following week, two exams per semester, and a compact one-semester track.
- Editing exercises: edit the notes (`notas/capNN_notas.tex`, section "Exercícios complementares") and regenerate the list with `gera_listas.py` — never edit `listas/listas_exercicios.tex` by hand.
- Adding a figure to a slide: `\figlivroslide{fig_NN_X.jpg}{height}{caption}` for a book figure (adds the CC attribution line); `\figlivroref{Fig.~N.M}{caption}{holder}` for a figure that is not redistributed (a boxed pointer to the book).

4. Tools reference (`ferramentas/`)

Every script has `--help`, `--version`, and the common options `-v/--verbose`, `-q/--quiet`, `--log-dir PASTA`. Each run writes a log under `<output folder>/logs/<script>-<UTC>.log` with the command line, the versions and the outcome (`RESULTADO: ok|FALHA`). Exit code 0 = success. Run them from anywhere; paths default to this repository.

Script	What	Options
`audita.py`	Consistency audit of the whole course (figures cited exist, no placeholders, cross-references, notebook pairs, 4 cases + 4 N per student notebook, 5 T + 4 N in solutions and notes, no error cells, guide complete). Exit 1 on any problem not in its accepted list.	`--root PASTA`, `--json ARQUIVO`
`gera_listas.py`	Generate `listas/listas_exercicios.tex` from the notes; `--pdf` also compiles it (two `pdflatex` passes).	`--notas PASTA`, `--outdir PASTA`, `--pdf`
`extract_plots.py`	Dump the `image/png` outputs of executed notebooks as PNG files (`<notebook>_plotNN.png`).	`--notebooks NOME ...`, `--pasta-notebooks PASTA`, `--outdir PASTA`
`contact_sheet.py NN`	Labelled thumbnail grid of `figuras/capNN/` to check the figure mapping by eye.	`--figuras PASTA`, `--out PNG`, `--cols N`, `--dpi N`

Script	What	Options
`render_qa.py`	Render sample pages of the compiled PDFs to PNG for visual QA (`--fase 1`: chapters 1–11 decks; `2`: chapters 12–14, notes, guide; `all`). Needs PyMuPDF.	`--root PASTA`, `--outdir PASTA`, `--fase {1,2,all}`, `--dpi N`
`slice.py PRIMEIRA ULTIMA NOME`	Cut a page range of the book PDF into `NOME.pdf` for the extractor (PDF page = book page + 16). Needs PyMuPDF and the book.	`--book PDF`, `--outdir PASTA`
`map_figs.py NN`	Map the extractor's crops for chapter NN to `figuras/capNN/fig_NN_X.jpg` (one caption, one figure; several images → suffixes a, b, c). Needs the extraction.	`--extracted PASTA`, `--figuras PASTA`

`docs/build_manual.py` builds this manual (`--outdir PASTA`, `--no-pdf`, `--verbose`); it uses pandoc + xelatex/lualatex when present and a built-in converter otherwise.

5. Quality checks

- `python ferramentas/audita.py` — the course-level audit above.
- `python -m unittest discover -s tests -v` — the 25-check suite: licence and disclaimer guards, SPDX headers, version consistency, the figure policy (section 7), notebook QA (46 notebooks, 23 pairs, no error or stderr output, every code cell executed, ≤ 1.5 MB each), the docs guard (every script flag appears in this manual and in `AGENTS.md`, README counts match the tree), read-back of every committed PDF with PyMuPDF (page counts and text), run logs, and the vendored publication-conformance checker.
- `python -m pyflakes ferramentas comum docs tests` — static check, also the first CI step. CI runs both on Linux, Windows and macOS (Python 3.11 and 3.13). CI has no TeX: it reads the committed PDFs, it does not rebuild them.
- After executing a notebook, scan its JSON for `output_type: error` and `stderr` streams — a clean exit code is not enough (the suite does this).

6. Every feature and every known limitation

Features (each one is verified by the suite or by the audit):

1. 23 chapters × 5 pieces, all present, all compiled/executed (46 notebooks).
2. Every book figure used on a slide exists in `figuras/` and carries the CC BY-NC-SA attribution line; non-redistributable figures are replaced by a boxed pointer to the book.
3. Exercise lists are generated from the notes, never hand-edited.
4. Both math forms side by side: the book's algebraic form and the differential form, in every chapter's notes.
5. Every chapter has at least one small numerical simulation ("previsão numérica em miniatura") and four worked cases with the full Python stack.
6. Solutions notebooks verify the theory exercises with SymPy.
7. Consistent plotting style (`comum/estilo_meteo.py`): course palette, shaded atmospheric layers, decimal commas in annotations.
8. Maintenance tools with complete command-line parameters and run logs.

Known limitations:

1. Language: everything a student sees is in Portuguese; only this manual, `AGENTS.md`, the README and the licence files are in English.
2. Non-commercial only for the teaching content (CC BY-NC-SA 4.0, inherited from the book); the code is Apache-2.0. See section 7.
3. 13 book figures are not included (24 crops: figs. 8.11, 8.16a/d, 9.20, 14.1, 14.3, 14.5, 15.3, 15.18, 16.19, 17.6, 20.15, 20.19, and 7.13 was never included) because the book credits them to third parties; six slides show the boxed pointer instead of the image.
4. The book is not redistributed; rebuilding figures needs your own download and an extractor. The shipped crops are as good as the extractor made them (vector figures rasterised at page resolution).
5. Chapter 23 (climate change) is not a chapter of Stull's book: its slides use notebook figures only; its content follows the notes.
6. Schedule dates are placeholders until the academic calendar is known.
7. Deliberate omissions at curiosity level (recorded in the audit of 2026-08-28): the diagram-identification quick guide (§5.4), SYNOP codes in detail (§9.2), MOS (§20.5), crepuscular rays (§22).
8. `audita.py` has two accepted findings: chapter 1 has five cases and states its N exercises in the notes rather than in the notebook (the approved template) — both are listed in the script's `EXCECOES`.
9. Notebook execution is not part of CI (the `meteo` environment is heavy); the committed notebooks are executed locally and the suite checks the committed outputs.
10. Windows LaTeX trap: a PDF open in a viewer locks `pdflatex`'s output; close the viewer or compile with `-jobname`.
11. Unicode in LaTeX: Greek letters, the approximately-equal sign or a check mark typed directly break `pdflatex`; use the LaTeX commands.
12. Third-party cross-checks are numerical, not refereed: values in the notebooks follow the book and standard references; teachers remain responsible for what they teach.

7. Licence and figure policy

- Code (`ferramentas/`, `comum/*.py`, `comum/*.sty`, `docs/*.py`, `tests/`, the code cells of the notebooks): Apache License 2.0 (`LICENSE`, `NOTICE`).
- Teaching content (notes, slides, guide, lists, index, figures, the text cells of the notebooks): derived from Stull's CC BY-NC-SA 4.0 book, therefore CC BY-NC-SA 4.0 (`LICENSES/CC-BY-NC-SA-4.0.txt`): credit the book and this project, no commercial use, share adaptations under the same licence.
- Figures: a figure the book credits to a third party ("courtesy of", "used with permission", a copyright line) is covered by that holder's permission to the book, not by the book's licence, and is not redistributed here. `tests/test_course.py` carries the banned list and fails if any of them reappears in `figuras/` or in a `.tex` file. US-government images (NOAA, NWS, NASA) are public domain and stay, with their credit in the slide caption.
- The warranty disclaimer and limitation of liability are in `LICENSE` (Apache §7–8), in the CC text (section 5) and in the README.

8. Layout

```
indice.tex/.pdf          course map - start here
notas/                   capNN_notas.tex/.pdf
slides/                  capNN_slides.tex/.pdf
guia_do_professor/       guia.tex (+ capNN_guia.tex), guia.pdf, cronograma.tex/.pdf
notebooks/               capNN_<tema>.ipynb, capNN_solucoes.ipynb
listas/                  listas_exercicios.tex/.pdf (generated)
figuras/capNN/           fig_NN_X.jpg (book), notebook_*.png (ours)
comum/                   preambulo.sty, temameteo.sty, estilo_meteo.py
ferramentas/             the tools of section 4 (+ registro.py, shared logging)
```

docs/	this manual (+ html/pdf), build_manual.py, DEVLOG.md, AUDIT-*.md
tests/	the suite + vendored conformance checker
book/, extracted/	your local copy of the book and its extraction (gitignored)